25MCB1010
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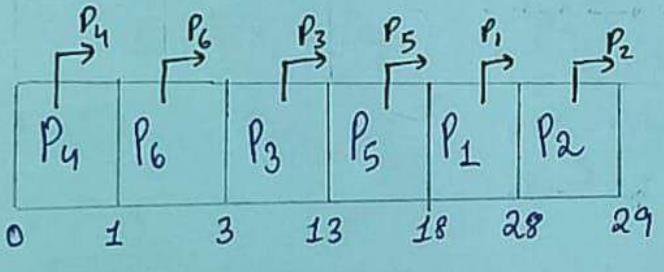
OS: Digital Assignment - 1

① First Come First Serve

Criterion: AT > P_{ID}

P_{ID}	AT	BT	CT	TAT	WT	RT
P_1	2	10	28	26	16	16
P_2	5	1	29	24	23	23
P_3	1	10	13	12	2	2
P_4	0	1	1	1	0	0
P_5	1	5	18	17	12	12
P_6	0	2	3	3	1	1

Gantt Chart:



Avg. TAT = 83/6 = 13.8

Avg. WT = 54/6 = 9.0

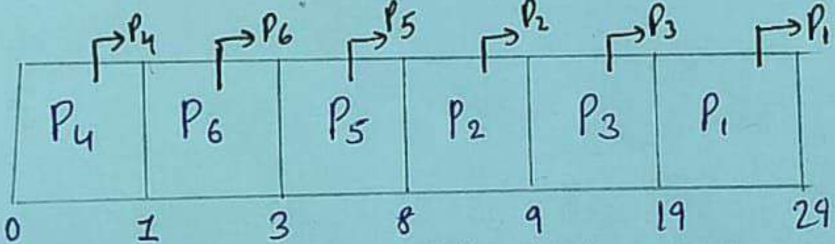
Avg. RT = 54/6 = 9.0

② Shortest Job first - Non-Premptive

Criterion : B.T > AT > P_{ID}

P_{ID}	AT	BT	CT	TAT	WT	RT
P_1	2	10	29	27	17	17
P_2	5	1	9	4	3	3
P_3	1	10	19	18	8	8
P_4	0	1	1	1	0	0
P_5	1	5	8	7	2	2
P_6	0	2	3	3	1	1

Gantt Chart :



0	1	3	8	9	19
~~$P_4(1)$~~	~~$P_6(2)$~~	$P_3(10)$	$P_3(10)$	~~$P_3(10)$~~	~~$P_1(10)$~~
$P_6(2)$	$P_3(10)$	$P_1(10)$	$P_1(10)$	$P_1(10)$	
	$P_5(5)$	~~$P_5(5)$~~	~~$P_2(1)$~~		

Avg. TAT = 60/6 = 10.0

Avg. WT = 31/6 = 5.16

Avg. RT = 31/6 = 5.16

③ Shortest Remaining Time first - Preemptive

Criterion : Remaining Burst Time > AT > P_{ID}

P_{ID}	AT	BT	CT	TAT	WT	RT
P_1	2	10	29	27	17	17
P_2	5	1	6	1	0	0
P_3	1	10	19	18	8	8
P_4	0	1	1	1	0	0
P_5	1	5	9	8	3	2
P_6	0	2	3	3	1	1

Gantt Chart:

$\to P_4$ $\to P_6$ $\to P_2$ $\to P_5$ $\to P_3$ $\to P_1$

P_4	P_6	P_6	P_5	P_5	P_2	P_5	P_3	P_1

0 1 2 3 4 5 6 9 19 29

0	1	2	3	4	5	6	9	19
~~$P_4(1)$~~	$P_6(2)$	~~$P_6(1)$~~	$P_3(10)$	$P_5(4)$	$P_5(3)$	~~$P_5(3)$~~	~~$P_3(10)$~~	~~$P_1(10)$~~
$P_6(2)$	$P_3(10)$	$P_3(10)$	$P_5(5)$	$P_3(10)$	$P_3(10)$	$P_3(10)$	$P_1(10)$	
	$P_5(5)$	$P_5(5)$	$P_1(10)$	$P_1(10)$	$P_1(10)$	$P_1(10)$		
		$P_1(10)$			~~$P_2(1)$~~			

Avg TAT = $58/6 = 9.66$

Avg WT = $29/6 = 4.83$

Avg RT = $28/6 = 4.66$

④ Priority - Non premptive

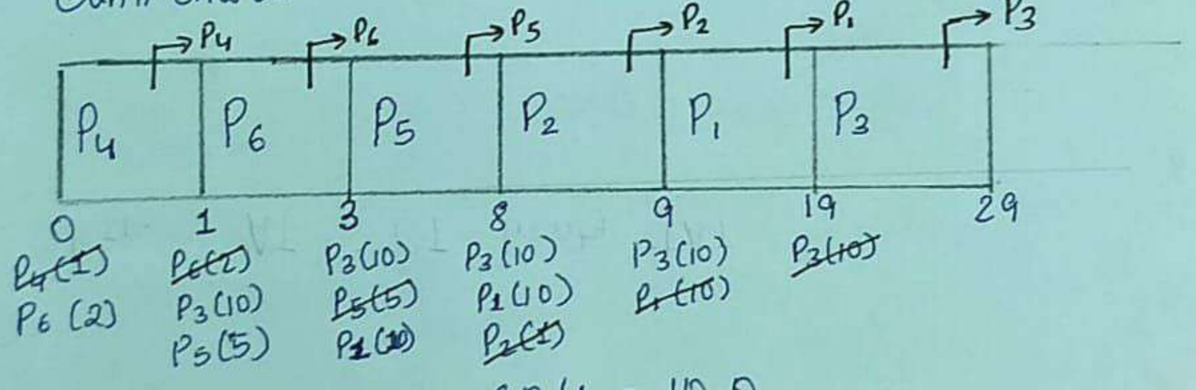
Criterion: Priority > Arrival Time > Process ID

Condition: Lower the Burst Time, Higher the Priority

P_{ID}	AT	BT	Priority	CT	TAT	WT	RT
P_1	2	10	5	19	17	7	7
P_2	5	1	1	9	4	3	3
P_3	1	10	6	29	28	18	18
P_4	0	1	2	1	1	0	0
P_5	1	5	4	8	7	2	2
P_6	0	2	3	3	3	1	1

Priority Range: $P_2 > P_4 > P_6 > P_5 > P_1 > P_3$

Gantt Chart:



Avg Turn Around Time = 60/6 = 10.0

Avg Waiting Time = 31/6 = 5.16

Avg Response Time = 31/6 = 5.16

⑤ Priority Premptive

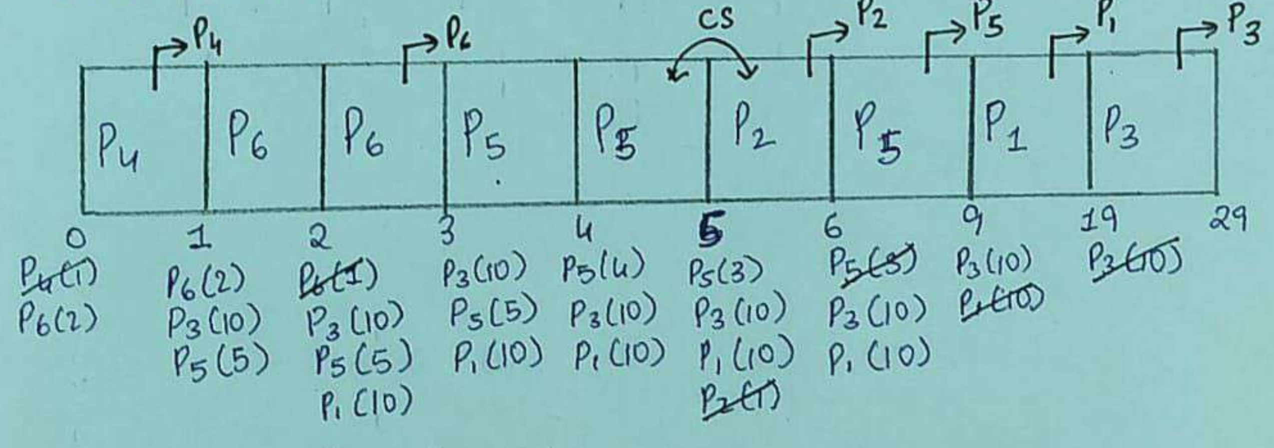
Criterion: Priority > Arrival Time > Process ID

Condition: Lower the Burst time, Higher the Priority

PID	AT	BT	Priority	CT	TAT	WT	RT
P_1	2	10	5	19	17	7	7
P_2	5	1	1	6	1	0	0
P_3	1	10	6	29	28	18	18
P_4	0	1	2	1	1	0	0
P_5	1	5	4	9	8	3	2
P_6	0	2	3	3	3	1	1

Priority Range: $P_2 > P_4 > P_6 > P_5 > P_1 > P_3$

Gantt Chart:



Avg. TAT = 58/6 = 9.66

Avg. WT = 29/6 = 4.83

Avg. RT = 28/6 = 4.66

⑥ Round Robin - Premptive

Time Quantum = 3

P_{ID}	AT	BT	CT	TAT	WT	RT
P_1	2	10	29	27	17	7
P_2	5	1	13	8	7	7
P_3	1	10	28	27	17	2
P_4	0	1	1	1	0	0
P_5	1	5	18	17	12	5
P_6	0	2	3	3	1	1

Gantt Chart:

→P_4 →P_6 →P_2 →P_5 →P_3 →P_1

P_4	P_6	P_3	P_5	P_1	P_2	P_3	P_5	P_1	P_3	P_1	P_3	P_1

0 1 3 6 9 12 13 16 18 21 24 27 28 29

Ready Queue:

~~P_4~~	~~P_6~~	P_3	P_5	P_1	~~P_2~~	P_3	~~P_5~~	P_1	P_3	P_1	~~P_3~~	~~P_1~~
1	2	~~10~~ 7	~~5~~ 2	~~10~~ 7	1	~~7~~ 4	2	~~7~~ 4	~~4~~ 1	~~4~~ 1	1	1

Avg. TAT = 83/6 = 13.83

Avg. WT = 54/6 = 9.0

Avg. RT = 22/6 = 3.66

Summary of Scheduling Results - FCFS / SJB / SRTF / Priority / RR

Results Table :

Algorithm	Avg. TAT	Avg. WT	Avg. RT
FCFS	13.8	9.0	9.0
SJF	10.0	5.16	5.16
SRTF	9.66	4.83	4.66
Priority (non-premptive)	10.0	5.16	5.16
Priority (premptive)	9.66	4.83	4.66
Round Robin	13.83	9.0	3.66

Definitions :

Turnaround time (TAT) : measures total time spent, time from process arrival to completion.

Waiting Time (WT) : total time spent by process waiting in the ready queue not considering execution.

Response Time (RT) : time taken from arrival till when the process gets CPU for first time.

Best Cases :

Best TAT : SRTF and Premptive Priority Scheduling (9.66)

Best WT : SRTF and Premptive Priority Scheduling (4.83)

Best RT : Round Robin (3.66)

If the goal is to minimize the overall completion and waitime times Shortest Remaining Time first and Premptive priority is suitable.

If the goal is to minimize first response latency Round Robin is suitable but at cost of TAT and WT.